

16.0 ELECTRICAL CHARACTERISTICS

Absolute Maximum Ratings

Storage temperature	-65°C to +150°C
Ambient temperature under bias	-40°C to +85°C (Industrial) 0°C to +70°C (Commercial)
Voltage on VDD, VDDOSC, VDDPLL, VDDR _X and VDDT _X , with respect to VSS	-0.3V to 3.6V
Voltage on $\overline{\text{RESET}}$, $\overline{\text{CS}}$, SCK and SI, with respect to VSS	-0.3V to 6.0V
Voltage on CLKOUT, SO, OSC1, OSC2, LEDA and LEDB, with respect to VSS	-0.3V to VDD + 0.3V
Voltage on TPIN+/- and TPOUT+/- with respect to VSS	-0.3V to 5.0V
VCAP with respect to VSS (Note 1)	-0.3V to 2.75V
ESD protection on all pins	2 kV
Current sourced or sunk by LEDA, LEDB	12 mA
Current sourced or sunk by CLKOUT	8 mA
Current sourced or sunk by $\overline{\text{INT}}$ and SO	4 mA

Note 1: VCAP is not designed to supply an external load. No external voltage should be applied to this pin.

† **Notice:** Stresses above those listed under "Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operational listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

ENC28J60

16.1 DC Characteristics: ENC28J60 (Industrial and Commercial)

DC CHARACTERISTICS			Standard Operating Conditions -40°C ≤ TA ≤ +85°C, 3.10V ≤ VDD ≤ 3.60V (Industrial) 0°C ≤ TA ≤ +70°C, 3.10V ≤ VDD ≤ 3.60V (Commercial)				
Param. No.	Sym	Characteristic	Min	Typ	Max	Units	Conditions
D001	VDD	Supply Voltage	3.10	3.30	3.60	V	
D002	VPOR	VDD Power-on Reset Voltage	—	—	0.7	V	See section on Power-on Reset for details
D003	SVDD	VDD Rise Rate to Ensure Internal Power-on Reset Signal	0.05	—	—	V/ms	See section on Power-on Reset for details
D004	VIH	Input High Voltage SCK, $\overline{\text{CS}}$, SI, $\overline{\text{RESET}}$	2.25	—	5.5	V	
D005		OSC1	0.7 VDD	—	VDD	V	
D006	VIL	Input Low Voltage SCK, $\overline{\text{CS}}$, SI, $\overline{\text{RESET}}$	VSS	—	1.0	V	
D007		OSC1	VSS	—	0.3 VDD	V	
	VOH	Output High Voltage LEDA, LEDB CLKOUT INT, SO	VDD - 0.7 VDD - 0.7 VDD - 0.7	— — —	— — —	V V V	IOH = -12.0 mA (Note 1) IOH = -8.0 mA (Note 1) IOH = -4.0 mA (Note 1)
	VOL	Output Low Voltage LEDA, LEDB CLKOUT INT, SO	— — —	— — —	0.4 0.4 0.4	V V V	IOL = 12.0 mA IOL = 8.0 mA IOL = 4.0 mA
	RPU	Weak Pull-up Resistance	74K	—	173K	Ω	
	IIL	Input Leakage Current All Input pins except OSC1	—	—	±1	μA	$\overline{\text{CS}} = \overline{\text{RESET}} = \text{VDD}$, VSS ≤ VPIN ≤ VDD, pins in high-impedance state (Note 1)
		OSC1 pin	—	—	±200	μA	OSC1 = VDD (Note 1)
	IDD	Operating Current Transmitting Ethernet Packets	—	160	180	mA	VDD = 3.30V, FSCK = 10 MHz, SO = Open, LEDA and LEDB Open, ECON2<PWRSV> = 0
		Active, not Transmitting Ethernet Packets	—	120	—	mA	VDD = 3.30V, LEDA and LEDB Open, ECON2<PWRSV> = 0
	IDDS	Standby Current (Sleep mode)	—	1.2	2.0	mA	$\overline{\text{CS}} = \text{VDD}$, Inputs tied to VDD or VSS, VDD = 3.3V, TA = 25°C, ECON2<PWRSV> = 1

Note 1: Negative current is defined as current sourced by the pin.

TABLE 16-1: AC CHARACTERISTICS: ENC28J60 (INDUSTRIAL AND COMMERCIAL)

AC CHARACTERISTICS	Standard Operating Conditions	
	$-40^{\circ}\text{C} \leq T_A \leq +85^{\circ}\text{C}$, $3.10\text{V} \leq V_{DD} \leq 3.60\text{V}$ (Industrial) $0^{\circ}\text{C} \leq T_A \leq +70^{\circ}\text{C}$, $3.10\text{V} \leq V_{DD} \leq 3.60\text{V}$ (Commercial)	

TABLE 16-2: OSCILLATOR TIMING CHARACTERISTICS

Param. No.	Sym	Characteristic	Min	Max	Units	Conditions
	FOSC	Clock In Frequency	25	25	MHz	
	TOSC	Clock In Period	40	40	ns	
	TDUTY	Duty Cycle (external clock input)	40	60	%	
	Δf	Clock Tolerance	—	± 50	ppm	

TABLE 16-3: RESET AC CHARACTERISTICS

Param. No.	Sym	Characteristic	Min	Max	Units	Conditions
	trl	$\overline{\text{RESET}}$ Pin High Time (between Reset events)	2	—	μs	
	trSTLOW	$\overline{\text{RESET}}$ Pin Low Time to Trigger Reset	400	—	ns	

TABLE 16-4: CLKOUT PIN AC CHARACTERISTICS

Param. No.	Sym	Characteristic	Min	Max	Units	Conditions
	$t_{h\text{CLKOUT}}$	CLKOUT Pin High Time	16.5	—	ns	TDUTY = 50% (Note 1)
	$t_{l\text{CLKOUT}}$	CLKOUT Pin Low Time	16.5	—	ns	TDUTY = 50% (Note 1)
	$t_{r\text{CLKOUT}}$	CLKOUT Pin Rise Time	—	3	ns	Measured from 0.1 VDD to 0.9 VDD, Load = 10 pF (Note 1)
	$t_{f\text{CLKOUT}}$	CLKOUT Pin Fall Time	—	4	ns	Measured from 0.9 VDD to 0.1 VDD, Load = 10 pF (Note 1)

Note 1: CLKOUT prescaler is set to divide by one.

TABLE 16-5: REQUIREMENTS FOR EXTERNAL MAGNETICS

Parameter	Min	Norm	Max	Units	Conditions
RX Transformer Turns Ratio	—	1:1	—	—	
TX Transformer Turns Ratio	—	1:1	—	—	Transformer Center Tap = 3.3V
Insertion Loss	0.0	0.6	1.1	dB	
Primary Inductance	350	—	—	μH	8 mA bias
Transformer Isolation	—	1.5	—	kV	
Differential to Common Mode Rejection	40	—	—	dB	0.1 to 10 MHz
Return Loss	-16	—	—	dB	

FIGURE 16-1: SPI INPUT TIMING

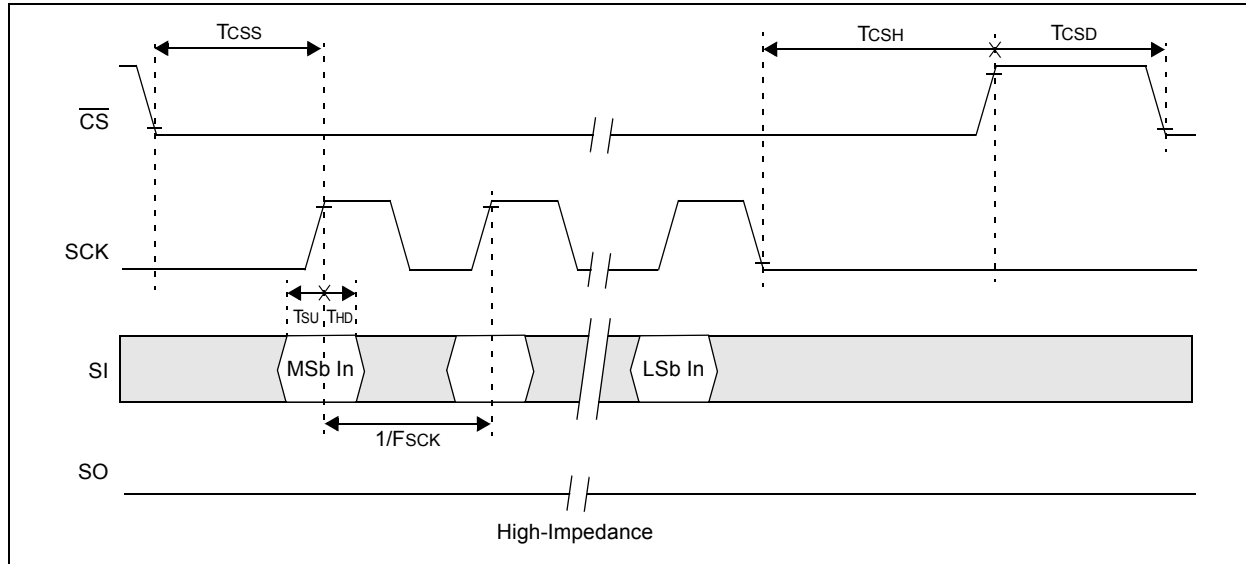


FIGURE 16-2: SPI OUTPUT TIMING

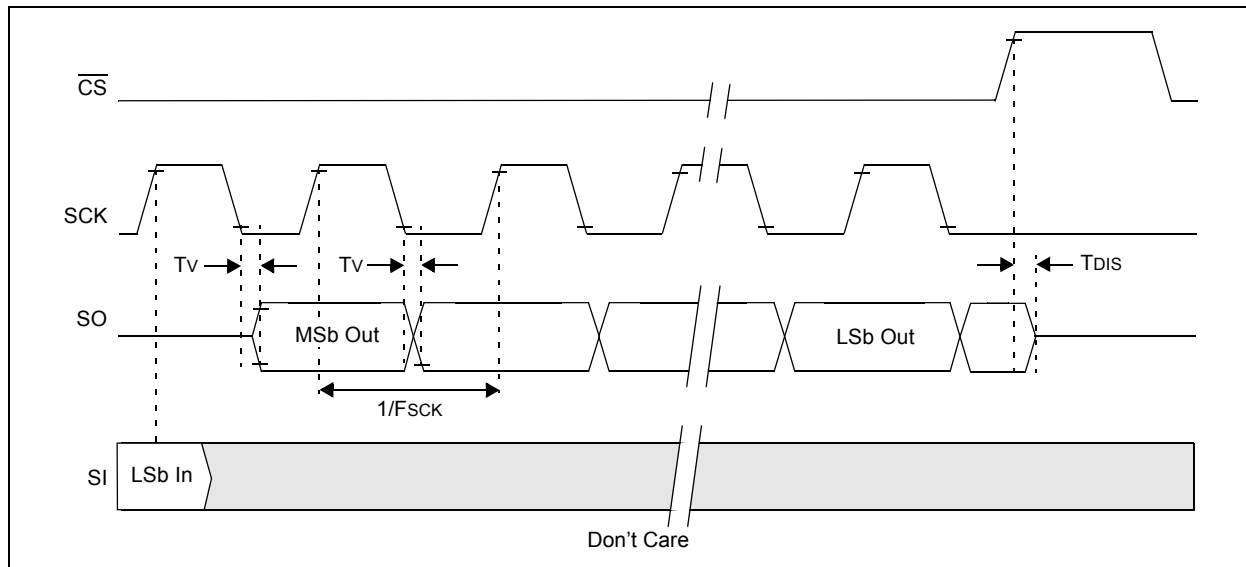


TABLE 16-6: SPI INTERFACE AC CHARACTERISTICS

Param. No.	Sym	Characteristic	Min	Max	Units	Conditions
	Fsck	Clock Frequency	DC	20	MHz	
1	Tcss	$\overline{CS}$ Setup Time	50	—	ns	
2	Tcsh	$\overline{CS}$ Hold Time	10	—	ns	ETH registers and memory buffer
			210	—	ns	MAC and MII registers
3	TcSD	$\overline{CS}$ Disable Time	50	—	ns	
4	TSU	Data Setup Time	10	—	ns	
5	THD	Data Hold Time	10	—	ns	
6	Tv	Output Valid from Clock Low	—	10	ns	SO Load = 30 pF
7	Tdis	Output Disable Time	—	10	ns	SO Load = 30 pF